LAB 3: ARCHITECTURAL PATTERN SELECTION & JUSTIFICATION

Course: Software Engineering (SELABS) | Student: **A R Akshay Kumar** | USN: **PES1UG24CS705**

Architecture Selection: *"We chose Layered Architecture for the Self-Service Coffee Kiosk System."*

1. Architectural Choice:

We selected the **3-Tier Layered Architecture** style (Presentation Layer, Business Layer, Data Layer). The Presentation Layer encapsulates the User Interface Component, handling customer touch screen interactions, menu browsing, and drink size selection. The Business Layer hosts the Order Manager Component (workflow orchestrator), Payment Service Component (card validation), and Receipt Printer Component (hardware abstraction). The Data Layer encapsulates the Database Component, maintaining menu catalog definitions, drink pricing rules, and order logs.

2. Two Scenario-Related Reasons for Selection:

- **Decoupling Hardware Peripherals & Device Drivers:** The kiosk interfaces directly with physical peripherals (touch screen display, thermal receipt printer) and financial hardware (credit card reader). Layered architecture isolates low-level hardware communications (ESC/POS printer serial commands) within dedicated components. Upgrading printer hardware or redesigning the touch screen UI requires zero changes to core order calculation and pricing logic.
- **Independent Menu, Pricing & Business Rule Administration:** Café menu items (Espresso, Americano, Latte) and size pricing multipliers (Small vs. Large) change frequently. By isolating persistence inside the Data Layer (Database Component), prices and recipes can be updated directly in the database without recompiling, redeploying, or risking regressions in kiosk UI workflows.

3. Security Advantage (PCI-DSS Cardholder Data Isolation):

Because the kiosk accepts credit card payments exclusively, securing payment data is essential. The Layered Architecture enforces strict layer boundaries: the touch UI layer never directly communicates with payment processing backends or internal database storage. The Payment Service Component resides in an isolated business layer boundary, transmitting Primary Account Numbers (PAN) strictly through PCI-DSS certified gateway APIs and masking account numbers. Unencrypted card numbers are never stored in the local database or logs, protecting customer financial data against physical terminal tampering.

4. Performance Benefit (In-Memory Caching & Sub-Second Responsiveness):

In a busy café with peak morning ordering spikes, kiosk latency directly influences customer queue times. In our Layered Architecture, the Order Manager and UI components load and cache menu catalog and pricing definitions in-memory upon kiosk startup. Customer touch actions (selecting coffee types, switching drink sizes, viewing order totals) are resolved instantaneously in-memory with sub-10ms response times without incurring disk I/O or network queries. Database writes are restricted to asynchronous transaction persistence upon payment completion, ensuring uninterrupted high customer throughput.